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SARDAR PATELUNIVERSITY
BCA (SEMESTER-2) Examination - 2023
US02CBCA21: Advanced C Programming

Date: 27/7/2023 Thursday
Time: 2:00p.m to 5:00pm

TOTAL MARKS: 70

NOTE: Figure to the right indicates full marks of the questions.

Q-1 MULTIPLE CHOICE QUESTIONS **[10]**

1. The variable declared inside a function is called ________
   a. Global b. Local c. function d. None of these

2. A function which calls itself is known as _______
   a. Reverse b. Recursive c. Reserve d. None of these

3. The parameter is also known as _____
   a. argument b. variable c. data type d. array

4. A ____________ is a collection of different data items.
   a. Array b. structure c. char d. none of these

5. Size of a union is determined by size of the.
   a.First member in the union b. Last member in the function
   c. Biggest member in the union d. Sum of the sizes of all members

6. The variables declared in a structure definition are called as its _______
   a. Objects b. Members c. Record d. None of these

7. Which of the following defines a pointer variable to an integer?
   a. int &ptr; b. int **ptr; c. int &&ptr; d. int *ptr;

8. Which of the following is not a C memory allocation function?
   a. malloc() b. calloc() c. realloc() d. alloc()

9. Which one of the following is valid for opening a file for only reading?
   a.fileOpen(filenm,"r"); b. fopen(filenm, "r");
   c.fileOpen(filenm, "ra") d. fileOpen(filenm, "read");

10. When fopen() is not able to open a file, it returns _________
    a. EOF b. NULL c. one d. zero

Q-2 Short Answer Questions (Attempt TEN out of TWELVE) **[20]**

1. Write and explain syntax for function call.
2. What is actual and formal parameter?
3. Which the advantages are of divide the program into functions?
4. Write and explain syntax for structure variable declaration.
5. Which elements are required for compile time initialization of a structure variable?
6. Differentiate structure and union.
7. Define pointer variable.
8. Define dynamic memory allocation.
9. What does free() do? Write the syntax for using free().
10. List basic file operations performed on file.
11. List out different file modes.
12. Differentiate: getc and getchar

(P.T.O.)

Q-3

[A] Explain Function declaration with syntax and example. [05]

[B] Explain function with no return type and no parameters. [05]

OR

Q-3

[A] Explain function with return type and no parameters. [05]

[B] Explain recursive function with example. [05]

Q-4

[A] Explain structure definition with syntax and example. [05]

[B] Explain array of structure with example. [05]

OR

Q-4

[A] Explain structure within structure with example. [05]

[B] Explain union with syntax and example. [05]

Q-5

[A] How can we access a variable through its pointer? Explain with proper example. [05]

[B] Explain calloc() with example. [05]

OR

Q-5

[A] Explain pointers as function arguments with example. [05]

[B] Explain malloc() with example. [05]

Q-6

[A] Explain fopen() in detail. [05]

[B] Explain getc() and putc() with example. [05]

OR

Q-6

[A] Explain fprintf() and fscanf() with example. [05]

[B] Explain getw() and putw() with example. [05]
